Omkar Sudhir Patil

Published Works

omkarsudhirpatil.com | omkarpatil64328@gmail.com | Google Scholar

Journal Articles

[1] **O. Patil**, R. Sun, S. Bhasin, and W. E. Dixon, "Adaptive control of time-varying parameter systems with asymptotic tracking," *IEEE Trans. Autom. Control*, vol. 67, no. 9, pp. 4809–4815, 2022.

[2] **O. Patil**, D. Le, M. Greene, and W. E. Dixon, "Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network," *IEEE Control Syst Lett.*, vol. 6, pp. 1855–1860, 2022.

[3] **O. Patil**, A. Isaly, B. Xian, and W. E. Dixon, "Exponential stability with RISE controllers," *IEEE Control Syst. Lett.*, vol. 6, pp. 1592–1597, 2022.

[4] **O. Patil**, R. Kamalapurkar, and W. Dixon, "A saturated RISE controller with exponential stability guarantees," *IEEE Trans. Autom. Control*, 2025.

[5] **O. Patil**, D. Le, E. Griffis, and W. Dixon, "Lyapunov-based deep residual neural network (ResNet) adaptive control," *IEEE Access*, 2025.

[6] **O. Patil**, B. C. Fallin, C. F. Nino, R. G. Hart, and W. E. Dixon, "Bounds on deep neural network partial derivatives with respect to parameters," *arXiv preprint arXiv:2503.21007*, 2025.

[7] A. Isaly, **O. Patil**, H. Sweatland, R. Sanfelice, and W. E. Dixon, "Adaptive safety with a RISE-based disturbance observer," *IEEE Trans. Autom. Control*, vol. 69, no. 7, pp. 4883–4890, 2024.

[8] E. Griffis, **O. Patil**, Z. Bell, and W. Dixon, "Lyapunov-based long short-term memory (Lb-LSTM) neural network-based control," *IEEE Control Syst. Lett.*, vol. 7, pp. 2976–2981, 2023.

[9] C. Nino, **O. Patil**, and W. Dixon, "Second-order heterogeneous multi-agent target tracking without relative velocities," *IEEE Control Systems Letters*, vol. 7, pp. 3663–3668, 2023.

[10] E. Griffis, **O. Patil**, R. Hart, and W. Dixon, "Lyapunov-based long short-term memory (Lb-LSTM) neural network-based adaptive observer," *IEEE Control Syst. Lett.*, vol. 8, pp. 97–102, 2024.

[11] R. Hart, E. Griffis, **O. Patil**, and W. Dixon, "Lyapunov-based physics-informed long short-term memory (LSTM) neural network-based adaptive control," *IEEE Control Syst. Lett.*, vol. 8, pp. 13–18, 2024.

[12] D. Le, **O. Patil**, C. Nino, and W. E. Dixon, "Accelerated gradient approach for deep neural network-based adaptive control of unknown nonlinear systems," *IEEE Trans. Neural Netw. Learn. Syst.*, 2025.

[13] C. F. Nino, **O. Patil**, C. D. Petersen, S. Phillips, and W. E. Dixon, "Collaborative spacecraft servicing under partial feedback using Lyapunov-based deep neural networks," *The Journal of the Astronautical Sciences*, 2025.

[14] C. Nino, **O. Patil**, J. Insinger, M. Eisman, and W. E. Dixon, "Online resnet-based adaptive control for nonlinear target tracking," *IEEE Control Syst. Lett.*, 2025.

[15] C. Nino, **O. Patil**, S. Edwards, and W. E. Dixon, "Distributed RISE-based control for exponential heterogeneous multi-agent target tracking of second-order nonlinear systems," *IEEE Control Syst. Lett.*, 2025.

[16] W. Makumi, **O. Patil**, and W. Dixon, "Lyapunov-based adaptive deep learning for approximate dynamic programming," *Automatica*, vol. 180, p. 112 462, 2025.

[17] E. Griffis, **O. Patil**, W. Makumi, and W. Dixon, "Adaptive output feedback control using lyapunov-based deep recurrent neural networks (Lb-DRNNs)," *IEEE Trans. Autom. Control*, 2025.

[18] R. Hart, **O. Patil**, Z. Bell, and W. Dixon, "Concurrent learning for system identification and control using lyapunov-based deep neural networks," *IEEE Control Syst. Lett.*, vol. 9, pp. 2957–2962, 2025.

Conference Papers

[19] **O. Patil**, D. M. Le, E. Griffis, and W. E. Dixon, "Deep residual neural network (ResNet)-based adaptive control: A Lyapunov-based approach," in *Proc. IEEE Conf. Decis. Control*, 2022.

[20] **O. Patil**, K. Stubbs, P. Amy, and W. E. Dixon, "Exponential stability with RISE controllers for uncertain nonlinear systems with unknown time-varying state delays," in *Proc. IEEE Conf. Decis. Control*, 2022.

[21] **O. Patil** and S. Bhasin, "Lyapunov based hierarchical trajectory control of an autonomous ground vehicle subjected to slip," in *IFAC World Congr.*, 2020.

[22] D. M. Le, **O. Patil**, C. Nino, and W. E. Dixon, "Accelerated gradient approach for neural network-based adaptive control of nonlinear systems," in *Proc. IEEE Conf. Decis. Control*, 2022.

[23] D. M. Le, **O. Patil**, P. Amy, and W. E. Dixon, "Integral concurrent learning-based accelerated gradient adaptive control of uncertain Euler-Lagrange systems," in *Proc. Am. Control Conf.*, Jun. 2022.

[24] A. Isaly, **O. Patil**, R. Sanfelice, and W. E. Dixon, "Adaptive safety with multiple barrier functions using integral concurrent learning," in *Proc. Am. Control Conf.*, 2021, pp. 3719–3724.

[25] R. Dasgupta, S. B. Roy, **O. Patil**, and S. Bhasin, "A singularity-free hierarchical nonlinear quad-rotorcraft control using saturation and barrier Lyapunov function," in *Proc. Am. Control Conf.*, 2019.

[26] E. Griffis, **O. Patil**, W. Makumi, and W. E. Dixon, "Deep recurrent neural network-based observer for uncertain nonlinear systems," in *IFAC World Congr.*, 2023.

[27] R. Hart, **O. Patil**, E. Griffis, and W. E. Dixon, "Lyapunov-based deep physics-informed neural network-based adaptive control," in *Proc. IEEE Conf. Decis. Control*, 2023.

[28] C. Nino, **O. Patil**, J. Philor, Z. Bell, and W. Dixon, "Deep adaptive indirect herding of multiple target agents with unknown interaction dynamics," in *Proc. IEEE Conf. Decis. Control*, 2023.

[29] C. F. Nino, **O. Patil**, S. C. Edwards, Z. I. Bell, and W. E. Dixon, "Distributed target tracking under partial feedback using Lyapunov-based deep neural networks," in *Proc. Am. Control Conf.*, 2025.

[30] C. F. Nino, **O. Patil**, C. D. Petersen, S. Phillips, and W. E. Dixon, "Collaborative spacecraft servicing under partial feedback using Lyapunov-based deep neural networks," in *Proc. AAS/AIAA Sp. Flight Mech. Meet.*, 2025.

[31] W. Wu, **O. Patil**, H. Sweatland, and W. Dixon, "Control contraction metric-based deep neural network adaptive control," in *Proc. IEEE Conf. Decis. Control*, 2025.

[32] S. Kolhe, **O. Patil**, J. Fu, and W. Dixon, "Integral concurrent learning control barrier functions for signal temporal logic tasks under unknown dynamics," in *Proc. IEEE Conf. Decis. Control*, 2025.